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INSTRUCTION TO CANDIDATES

ANSWER **ALL** QUESTIONS IN THE QUESTION PAPER.

(Student are required to write his/her name and lecturer's name on the answer script)

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1.0 Introduction and Problem Significance

Waste management is a critical environmental challenge faced by modern societies. Inefficient waste segregation leads to contamination of recyclable materials, reduced recycling efficiency, and increased environmental pollution. Certain waste categories such as batteries require special handling due to hazardous chemical components, while materials like glass and cardboard must be processed differently during recycling.

Manual waste sorting is labor-intensive and prone to human error, especially when dealing with large volumes of waste. With recent advancements in artificial intelligence, computer vision techniques such as Convolutional Neural Networks (CNNs) can be utilized to automate waste classification based on visual information. CNNs are well suited for image-based tasks as they are capable of learning hierarchical visual features directly from raw pixel data.

This project focuses on the development of an image classification system that categorizes garbage images into three classes: battery, brown glass, and cardboard. Two CNN-based approaches were implemented and compared: a custom-designed lightweight CNN and a transfer learning model using a pre-trained EfficientNet architecture. The objective is to evaluate their effectiveness and suitability for supporting environmentally sustainable waste management systems.

2.0 Dataset Overview & Preprocessing

2.1 Dataset Overview

The dataset used in this project consists of 2,443 grayscale images distributed across three garbage categories: battery, brown glass, and cardboard. Images are organized into class-specific folders, allowing labels to be automatically inferred during data loading. The dataset contains variations in object orientation, lighting conditions, background clutter, and surface textures.

The total dataset was internally split into training, validation, and testing subsets using a 70:15:15 ratio, resulting in:

- Training set: 1,710 images
- Validation set: 366 images
- Test set: 367 images

This splitting strategy ensures sufficient data for model learning while reserving a held-out test set for unbiased evaluation.

2.2 Preprocessing and Augmentation

All images were resized to 224×224 pixels to ensure consistency across inputs. Pixel values were normalized to the range $[0, 1]$ to improve training stability. To enhance generalization and reduce overfitting, data augmentation techniques were applied during training, including random horizontal flipping, rotation, brightness adjustment, and zooming.

3.0 Model Architectures & Training Details

This section explains the architectural design of the two models used in this project: a custom Convolutional Neural Network (CNN) trained from scratch and a transfer learning model based on EfficientNetB0. The design choices are justified in terms of feature learning capability, computational efficiency, and suitability for the given dataset.

3.1 Custom CNN Architecture

The custom CNN was designed as a lightweight architecture tailored for grayscale image classification. The input to the network is a $224 \times 224 \times 1$ image, where the single channel represents grayscale intensity. This design reduces computational complexity while retaining essential structural information such as edges, textures, and object shapes.

i. Convolutional Feature Extraction

The model consists of three convolutional blocks with increasing filter sizes:

- The first convolution layer (32 filters) captures low-level features such as edges and simple textures.
- The second and third convolution layers (64 filters) learn more complex spatial patterns and class-specific visual characteristics.

Each convolution layer uses a 3×3 kernel, which is a common choice in CNN design as it balances expressive power and computational efficiency. The ReLU activation function introduces non-linearity, enabling the model to learn complex decision boundaries.

ii. Spatial Downsampling with Max Pooling

After each convolution layer, MaxPooling is applied to reduce the spatial dimensions of feature maps. This operation:

- Decreases computational cost
- Improves robustness to small translations in object position

- Helps the model focus on the most salient features

Progressive downsampling allows the network to learn hierarchical representations, from local details to more global structures.

Regularization with Dropout

Dropout layers with a rate of 0.3 are introduced before and after the fully connected layers. Dropout randomly disables neurons during training, preventing co-adaptation of features and reducing overfitting. This is particularly important because the dataset size is moderate and the model is trained from scratch.

iii. Fully Connected Classification Head

The extracted feature maps are flattened and passed through a dense layer with 64 neurons, which combines learned spatial features into a compact representation. The final layer uses softmax activation to output probability scores for the three waste classes. The class with the highest probability is selected as the predicted label.

Overall, the custom CNN provides a baseline model that demonstrates how a manually designed architecture can learn visual features directly from data. However, its performance is limited by the amount of available training data and the complexity of inter-class visual similarities.

3.2 Transfer Learning Architecture Using EfficientNetB0

To overcome the limitations of training from scratch, a transfer learning approach was adopted using EfficientNetB0 as the backbone network. EfficientNetB0 is a state-of-the-art CNN architecture that achieves high accuracy while maintaining computational efficiency.

i. Rationale for Choosing EfficientNetB0

EfficientNetB0 was selected for several reasons:

- It uses compound scaling, which systematically balances network depth, width, and input resolution.
- It offers an excellent accuracy-to-parameter ratio, making it suitable for practical deployment.
- Pre-trained weights on the ImageNet dataset provide strong general-purpose visual representations.

These characteristics make EfficientNetB0 particularly effective when working with datasets that are smaller than ImageNet.

ii. Feature Extraction with a Frozen Backbone

In this project, EfficientNetB0 was used as a feature extractor. The pre-trained convolutional layers were frozen to preserve learned representations and prevent overfitting. Freezing the backbone also significantly reduces training time and computational cost. The input images are passed through the EfficientNetB0 backbone, which extracts rich hierarchical features such as object contours, textures, and higher-level semantic patterns. These features are far more discriminative than those learned by a shallow network trained from scratch.

iii. Global Average Pooling and Classification Head

Instead of flattening feature maps, Global Average Pooling (GAP) is used to aggregate spatial information. GAP reduces the number of trainable parameters and encourages the network to focus on the presence of features rather than their exact location. This improves generalization and robustness. A dropout layer is applied after pooling to further reduce overfitting. The final dense layer with softmax activation produces probability scores for the three waste classes.

3.3 Architectural Comparison and Rationale for EfficientNetB0 Selection

The primary distinction between the two architectures lies in how visual features are learned and represented. The custom CNN learns all feature representations directly from the training dataset. While this approach provides flexibility and simplicity, it strongly depends on dataset size and diversity. With a moderate number of training images and visually similar classes, the custom CNN is more susceptible to overfitting and confusion between categories such as battery and brown glass.

In contrast, the transfer learning model employs EfficientNetB0, which leverages feature representations learned from the large-scale ImageNet dataset. These pre-trained features capture a wide range of visual primitives, including edges, textures, object parts, and global shape information. As a result, the transfer learning model starts with a much stronger feature foundation and requires fewer task-specific adjustments to achieve high performance.

EfficientNetB0 was selected over other popular architectures such as VGG, ResNet, or Inception for several key reasons. First, EfficientNetB0 offers an excellent accuracy-to-parameter efficiency, achieving competitive performance with significantly fewer parameters. This makes it computationally efficient and well suited for deployment in resource-constrained environments, such as embedded systems used in automated waste sorting facilities. Second, EfficientNet uses compound scaling, which systematically balances network depth, width, and input resolution. This design

leads to better feature utilization and improved generalization compared to architectures that scale these dimensions independently.

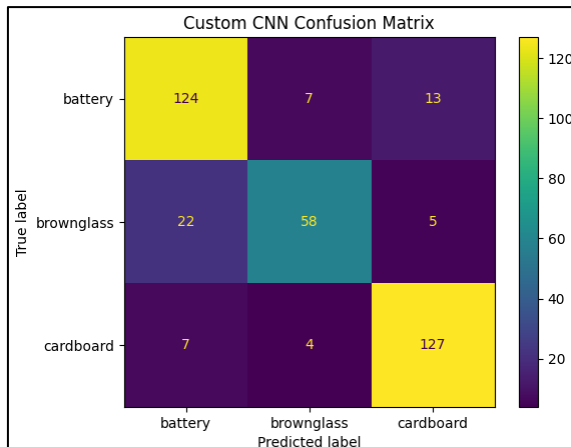
Furthermore, EfficientNetB0 operates natively at an input resolution of 224×224 , which aligns well with the preprocessing pipeline used in this project. This avoids unnecessary rescaling or architectural modifications that might be required for larger models. Compared to deeper variants of EfficientNet, the B0 version provides a good trade-off between performance and training stability, especially when the available dataset is not extremely large.

The use of Global Average Pooling in the transfer learning architecture further enhances generalization by reducing the number of trainable parameters and encouraging the model to focus on feature presence rather than precise spatial location. Freezing the EfficientNet backbone ensures that the learned representations remain stable while the newly added classification head adapts to the waste classification task.

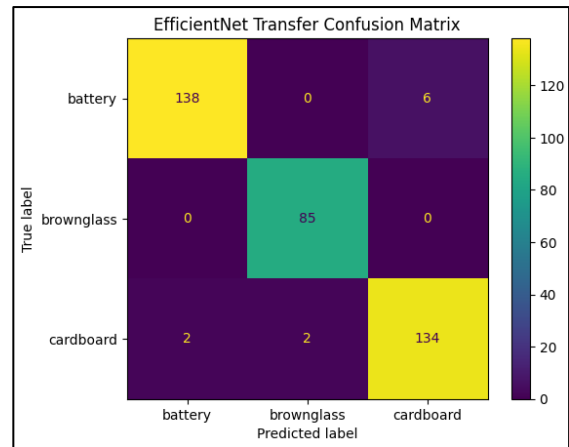
Overall, this architectural comparison demonstrates that while the custom CNN serves as a useful baseline for understanding feature learning from scratch, the EfficientNetB0-based transfer learning model is more robust, efficient, and scalable. Its design choices and pre-trained knowledge enable superior classification performance, making it a more suitable candidate for real-world waste classification systems.

4.0 Results with Visualizations

This section presents the performance evaluation of the two implemented models: the custom Convolutional Neural Network (CNN) and the transfer learning model based on EfficientNet. Model performance was assessed using standard classification metrics, namely precision, recall, F1-score, and overall accuracy, computed on a held-out test dataset. Confusion matrices were also generated to visualize classification behavior across classes.



The custom CNN achieved an overall accuracy of 0.8420. For the battery class, precision, recall, and F1-score were 0.8105, 0.8611, and 0.8350, respectively. The brown glass class recorded a precision of 0.8406, recall of 0.6824, and F1-score of 0.7532, indicating weaker performance due to visual similarity with battery images. The cardboard class achieved the highest performance, with precision 0.8759, recall 0.9203, and F1-score 0.8975. The confusion matrix shows most misclassifications occurred between battery and brown glass.



The transfer learning model significantly outperformed the custom CNN, achieving an overall accuracy of 0.9728. For the battery class, precision, recall, and F1-score were 0.9857, 0.9583, and 0.9718. The brown glass class achieved near-perfect performance with precision 0.9770, recall 1.0000, and F1-score 0.9884. The cardboard class also performed strongly, with precision 0.9571, recall 0.9710, and F1-score 0.9640. The confusion matrix indicates minimal misclassification across all classes.

A direct comparison between the two models reveals that the transfer learning approach outperforms the custom CNN across all evaluation metrics. While the custom CNN achieved acceptable performance, particularly for the cardboard class, it struggled with visually similar categories such as battery and brown glass. In contrast, the transfer learning model achieved high precision, recall, and F1-scores for all classes, demonstrating superior generalization capability.

These results indicate that transfer learning is a more suitable approach for garbage classification tasks, especially when working with limited datasets and subtle inter-class visual differences.

5.0 Ethical and Practical Reflections

From a practical perspective, the developed system demonstrates strong potential for deployment in automated waste sorting facilities. Accurate classification of waste materials can reduce contamination, improve recycling efficiency, and lower operational costs.

However, ethical considerations must be addressed. Dataset bias may limit generalization to real-world conditions where lighting, background, and object appearance differ significantly. Over-reliance on automated systems without human supervision could result in incorrect sorting decisions. Therefore, such systems should complement, rather than replace, human oversight.

Privacy concerns are minimal, as the model processes images of objects rather than individuals. Future work should focus on expanding dataset diversity and conducting real-world testing to ensure fair and reliable deployment.

6.0 Conclusion

This project successfully demonstrated the application of CNN-based image classification for garbage sorting. A comparison between a custom CNN and a transfer learning approach highlighted the advantages of leveraging pre-trained models, with EfficientNet achieving superior accuracy and generalization performance.

The results indicate that deep learning-based computer vision systems can play a significant role in supporting sustainable waste management initiatives. With further dataset expansion and real-world validation, such systems have the potential to contribute meaningfully to environmental conservation efforts.

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